

Choosing the Right Comparison Tool

Module 3, Part 1 (October 17)

QBA 5330

Where We're Headed Today

- Part 1: Choosing the right comparison tool
- Part 2: Working the Google Project Oxygen case
- Part 3: Team report-outs, synthesis, and what's next

Yesterday vs. Today

- **Friday:** quantifying uncertainty for *one* estimate (a mean, a proportion)
- **Today:** *comparing groups*, and asking whether a comparison supports a *causal* claim

Your Toolkit So Far

- One-sample tests
- Two-sample independent tests (pooled / Welch)
- Paired comparisons
- ANOVA and Tukey's HSD
- *(All already in your reading – today applies it.)*

Activity: Which Test Would You Use?

one-sample, two-sample independent, paired, or ANOVA?

Debrief

For each scenario: what would go wrong if you picked the wrong tool?

Recap: From Association to Causation

1. Was there **random assignment**?
2. Is there a **plausible confounder**?
3. Could **bias** explain the pattern instead of a real effect?
4. Is the effect **statistically significant**, and is it **practically significant**?

Today's Case

Google's people analytics team compared high-scoring and low-scoring managers' teams on satisfaction, retention, and performance.

Does better management *cause* better outcomes?

Before the Break

In Part 2, your job: apply everything from this hour to a real case.

- Which comparison did Google effectively make?
- Does the causation checklist support their conclusion?